

Q1:

a. +Z

b. +X

c. when $|\vec{E}| = 0 \Rightarrow \phi = \frac{\pi}{2}$

$$\vec{E} = E_0 \cos \left[(2\pi \times 10^6)t - \left(\frac{\pi}{150}\right)z + \phi^+ \right]$$

$= \frac{\pi}{2}$

$$\therefore \frac{\pi}{2} = (2\pi \times 10^6)(0.25 \times 10^{-6}) - \left(\frac{\pi}{150}\right)37.5 + \phi^+$$

$$\frac{\pi}{2} = \frac{\pi}{2} - \frac{\pi}{4} + \phi^+$$

$$\phi^+ = \frac{\pi}{4} \text{ rad}$$

d. $\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{150} \text{ rad/m}$

e.

$$\vec{E}_r(x, t) = E_0 \cos \left[(2\pi \times 10^6)t + \left(\frac{\pi}{150}\right)z + \frac{\pi}{4} \right] (-\vec{a}_x)$$

changing the
direction of
propagation

changing the
direction of
electric field

Q2:

$$a. \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (\text{or}) \quad \nabla \times \vec{E} = -j\omega \vec{B} \quad (\text{or}) \quad \nabla \times \vec{E} = -j\omega \mu \vec{H}$$

$$\vec{E} = \left[\frac{\theta e^{-j2\pi r}}{50r} \right] \vec{a}_\theta$$

$$E_r = 0$$

$$E_\theta = \frac{\theta e^{-j2\pi r}}{50r}$$

$$E_\phi = 0$$

$$\begin{aligned} \therefore -j\omega \mu \vec{H} &= \vec{a}_r \left(\frac{1}{r \sin \theta} \right) \left[-\frac{\partial E_\theta}{\partial \theta} \right] \\ &+ \vec{a}_\theta \left(\frac{1}{r} \right) [0] \\ &+ \vec{a}_\phi \left(\frac{1}{r} \right) \left[\frac{\partial}{\partial r} (r E_\theta) \right] \\ &= \vec{a}_\phi \left(\frac{1}{r} \right) \frac{\partial}{\partial r} \left[r \left(\frac{\theta e^{-j2\pi r}}{50r} \right) \right] \\ &= \vec{a}_\phi \left(\frac{1}{r} \right) \frac{\partial}{\partial r} \left[\frac{\theta e^{-j2\pi r}}{50} \right] \end{aligned}$$

$$-j\omega \mu \vec{H} = \vec{a}_\phi \left[\frac{1}{50r} \right] (\theta e^{-j2\pi r}) (-j2\pi)$$

$$\vec{H} = \vec{a}_\phi \left[\frac{2\pi}{50\omega \mu r} \right] (\theta e^{-j2\pi r})$$

[2]

$$\vec{H}(r, \theta, t) = \text{Re} \left\{ \vec{a}_\phi \left[\frac{2\pi}{50 \omega \mu r} \right] (\theta e^{-j2\pi r}) \right\}$$

$$\vec{H}(r, \theta, t) = \vec{a}_\phi \frac{2\pi \theta}{50 \omega \mu r} \cos[\omega t - 2\pi r] \quad \text{A/m}$$

b.

$$\eta_0 = \frac{E_0}{H_0} = \frac{\left(\frac{1}{50r} \right) \theta \cos[\omega t - 2\pi r]}{\left(\frac{2\pi \theta}{50 \omega \mu r} \right) \cos[\omega t - 2\pi r]}$$

$$\eta_0 = \frac{\omega \mu}{2\pi}$$

$$\lambda = 1 \text{ m}$$

since

$$\lambda = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi \times 3 \times 10^8}{6\pi \times 10^8}$$

$$\lambda = 1 \text{ m}$$

لذلك ظهرت معادلة η_0

$$\eta_0 = \frac{\omega \mu}{2\pi} \quad \text{بشكل}$$

$$\eta_0 = \frac{\omega \mu}{k} \quad \text{وليس}$$

Q3;

a.

$$\phi = \int \vec{B} \cdot d\vec{S}$$

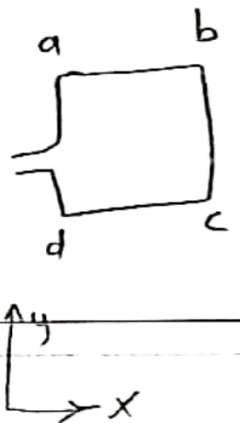
$$= \int [200x^2 \cos 10t] \vec{a}_z \cdot [dx dy] \vec{a}_z$$

$$= 200 \cos 10t \left[\int x^2 dx \int dy \right]$$

$$= 200 \cos 10t \left[\frac{x^3}{3} \right]_{+\frac{10}{100}}^{-\frac{10}{100}} \left[y \right]_{+\frac{10}{100}}^{-\frac{10}{100}}$$

$$= 200 \cos 10t \left[\frac{1}{3} (0.002) \right] [0.2]$$

$$= 0.0266 \cos 10t$$



$$V = -N \frac{\partial \phi}{\partial t} = -10 (0.0266) (-\sin 10t) (10)$$

$$= 2.66 \sin 10t \quad V$$

b. $V = -N \frac{\partial \phi}{\partial t}$

$$6.12 = -N (0.0266) (-\sin 10t) (10)$$

$$6.12 = N (0.266)$$

$$N = 23 \text{ turn}$$

$$\therefore \text{the extra-turns} = 23 - 10 = 13 \text{ turn}$$

[4]

Q4:

a.
$$V_{ab} = \int (\vec{U}_p \times \vec{B}) \cdot d\vec{L}$$

$$\vec{U}_p = \rho \omega \vec{a}_\phi = \frac{2\pi \times 96^3}{360} \rho \vec{a}_\phi = 3\pi \rho \vec{a}_\phi \quad \text{m/sec}$$

$$\vec{B} = [2 \times 10^{-4}] \vec{a}_z \quad \text{Wb/m}^2$$

$$\vec{U}_p \times \vec{B} = \begin{vmatrix} \vec{a}_\rho & \vec{a}_\phi & \vec{a}_z \\ 0 & 3\pi\rho & 0 \\ 0 & 0 & [2 \times 10^{-4}] \end{vmatrix}$$

$$= \vec{a}_\rho [6\pi \times 10^{-4}] \rho$$

$$V_{ab} = \int \rho [6\pi \times 10^{-4}] \vec{a}_\rho \cdot d\rho \vec{a}_\rho$$

$$= 6\pi \times 10^{-4} \left[\int_b^a \rho d\rho \right] = 6\pi \times 10^{-4} \left[\frac{\rho^2}{2} \right]_b^a$$

$$= 3\pi \times 10^{-4} [a^2 - b^2]$$

$$V_{ab} = -3\pi \times 10^{-4} b^2 \Big|_{b=0.5} = -7.5\pi \times 10^{-4} \quad \text{V}$$

b.
$$V_{ab} = -3\pi \times 10^{-4} b^2 \Rightarrow -12\pi \times 10^{-4} = -3\pi \times 10^{-4} b^2$$

$$\therefore b^2 = 4 \Rightarrow b = 2 \text{ m} \quad \text{the new rod length}$$